Supplementary material

Subject-independent surface EMG electrode reduction using cross-fitted attribution: a prespecified non-inferiority study

Supplementary methods S1–S6 are followed by supplementary figures S1–S2 and supplementary tables S1–S10, numbered in the order in which they are cited in the main text. Main-text tables are referred to as main table 1–3.

S1 Departures from the analysis plan

The following changes were made after the statistical analysis plan was written. None altered a prespecified decision rule (margin, statistical unit, bootstrap or definition of k^*).

1. The plan required Holm correction within H1. It was applied after the primary analysis and left k^* unchanged at all three margins (supplementary table S3). A fixed-sequence analysis, testing $k = 15, 14, \ldots$ and stopping at the first failure, is reported in the same table as a sensitivity analysis.

2. Training-sample sizes (40,000 windows for the support vector machine, 200,000 for the random forest and for the electrode-selection analyses) were chosen from held-out Macro-F1 on two folds (participants 1 and 7) before the main analyses. For these two participants the held-out data therefore informed a design choice.

3. The plan did not specify a criterion for choosing λ, so none was chosen and all four values are reported. The value $\lambda = 0.5$ used for SR-CFA in the H2 comparison was not chosen by a stated criterion, so the $\lambda = 0$ ranking is reported alongside it.

4. The plan named H1–H4 as confirmatory hypotheses with Holm correction within each family. The detailed H2–H4 test families (tests, rankings compared and subset sizes) were specified after the performance-curve and selection-method results were available; H2–H4 are therefore reported as secondary tests.

5. The comparison model for H2 and H4 was reduced from 150 to 50 trees for computational cost before any comparison result was available.

6. An initial count-feature threshold 87 times smaller than the final GRABMyo value saturated WAMP and was replaced. The leave-one-out check of the thresholds (section S2) was added after the primary analysis.

7. An initial deletion analysis combined deletion orders computed with different training sizes; it was replaced by the analysis reported here, in which all orders share one configuration (section S3).

8. The bipolar analysis was run three times. The first run used the monopolar count-feature threshold; the second used 5×10^{-3} mV, whose stated derivation could not be reproduced; the reported analysis uses 4×10^{-3} mV, the value given by the documented rule. Earlier outputs are archived.

9. The fixed-layout and feature-redundancy analyses (sections S4 and S5) were not in the plan. The fixed-layout check used $\lambda = 0.5$; no rationale for this value was recorded (section S4).

10. The participant-level attribution extension (section S6) was specified after the original results were available.

11. The window-length sensitivity analysis and the 17-class task, listed as exploratory in the plan, were not run.

S2 Leave-one-out check of the count-feature thresholds

All thresholds follow one rule: 1.15 times the median absolute first difference of the signal, rounded to one significant figure. The GRABMyo threshold was derived from 54 unfiltered recordings of 6 participants. For every outer fold we recomputed the median without the held-out participant and applied the same rule. It gave 10^{-2} mV in 43 of 43 folds (leave-one-out medians 8.52–9.45×10^{-3} mV); only 6 folds held out a participant who contributed to the sample. For the derived bipolar signal

(median 3.40×10^{-3} mV on the same recordings) it gave 4×10^{-3} mV in 43 of 43 folds. For DB5 it gave 7 quantisation steps in 10 of 10 folds. Deriving the thresholds within each fold would therefore have produced identical features. Because the thresholds were not estimated within folds, we describe this as a verification rather than as fold-local estimation.

S3 Consistency of the deletion analysis

The Remove-Bottom subsets coincide with the H2 cross-fitted SHAP subsets in all 172 participant-by-size cells, and their Macro-F1 values agree to within a mean absolute difference of 0.0014 (maximum 0.0084). The small differences arise because features were standardised separately for each subset in the deletion analysis and once over all features in the H2 comparison.

S4 Fixed-layout consistency check

A single 12-electrode layout was derived from a cross-fitted ranking on the 22 odd-numbered participants, removing electrodes 3, 9, 11, 12. Models with 16 and with 12 electrodes were trained on those participants and tested on the 21 even-numbered participants. The 12-electrode model scored -0.0124 against the 16-electrode model (95% confidence interval -0.0179 to -0.0077), within the margin. The electrode count was chosen after inspecting the full-cohort performance curve, which included the even-numbered participants. Although electrode identities were selected using development participants only, this is therefore not an independent confirmation of the chosen count. It is a single-split consistency check and does not constitute a prospective validation of a fixed sparse hardware layout.

The ranking for this check used $\lambda = 0.5$, whereas the primary analysis uses $\lambda = 0$; no rationale for this choice was recorded. In the primary LOSO analysis the $\lambda = 0.5$ and $\lambda = 0$ rankings selected a different top-12 set in 0 of 43 outer folds, but this was not verified for the development split used here. The removed set differs by one electrode from the set excluded in the primary outer folds (9, 11, 12, 16). [AUTHOR CHECK REQUIRED: justify the use of $\lambda = 0.5$ in the fixed-layout check, or repeat the check with $\lambda = 0$.]

S5 Feature redundancy

Integrated EMG equals 512 times the mean absolute value for these windows. Repeating the comparison-model evaluation without integrated EMG changed Macro-F1 by -0.0005 (95% confidence interval -0.0021 to $+0.0010$; 43 folds). The median per-fold Spearman ρ between rankings with and without integrated EMG was 0.994, and the top-12 set was identical in 34 of 43 folds, so the duplicated feature had little measurable effect on the ranking.

S6 Participant-level attribution extension

The post-hoc extension uses the same out-of-fold attributions as the original ranking, but gives each outer-training participant equal weight before computing the median and median absolute deviation. All penalties and subset sizes were evaluated with the same training budget as the original curve. Every reused fit records its source, and inputs are fingerprinted to prevent mixing incompatible runs. Between-person dispersion is descriptive heterogeneity, not a standard error of attribution. It includes both participant differences and differences between the inner fitted models.

Participant-level attribution extension (post hoc) All 43 outer participants were evaluated under each aggregation. The descriptive median relative MAD across outer-fold/electrode cells increased from 2.40% to 4.62%; these cells are not independent replicates. Of the 60 reduced-size, penalty-specific comparisons with the original fold aggregation, 42 had any nonzero participant difference. After Holm correction across this family, 0 favoured participant aggregation and 0 favoured fold aggregation. The full set of penalties is reported in the supplementary material; no candidate was selected using outer-test performance. These post-hoc comparisons test the aggregation change and do not replace the original H3 result. Within participant aggregation, 0 of 12 Holm-corrected comparisons supported a benefit of the stability penalty and 0 supported harm.

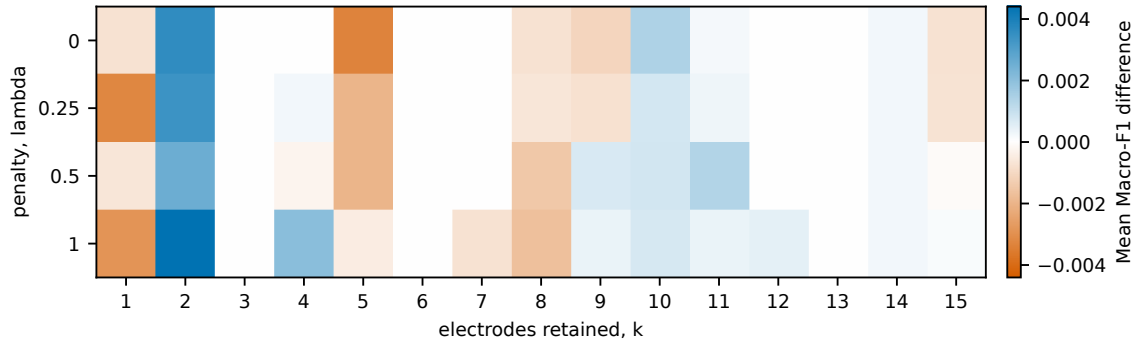


Figure S1. Participant-level versus fold-level attribution aggregation (post hoc; GRABMyo, 43 participants). Every cell is the mean paired Macro-F1 difference across participants, comparing person-level with fold-level aggregation under the same penalty, subset size and model protocol. Blue favours person-level aggregation and orange favours fold aggregation. Asterisks identify Holm-corrected differences within the complete family. All candidates are shown; outer-test outcomes did not select a penalty.

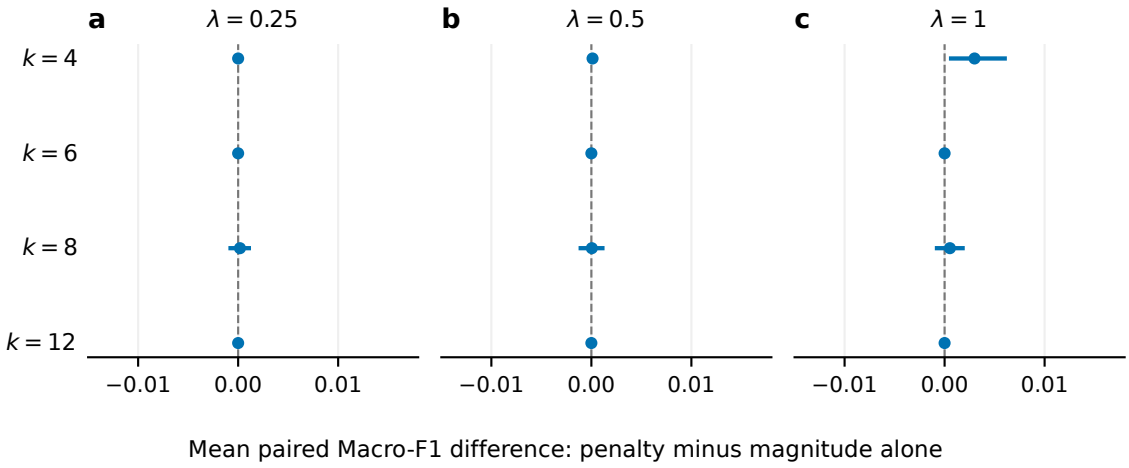


Figure S2. Stability penalty versus attribution magnitude alone under participant-level aggregation (post hoc). (a) $\lambda = 0.25$, (b) $\lambda = 0.5$, (c) $\lambda = 1$. Each point compares a positive penalty with magnitude alone. Bars are unadjusted participant-bootstrap 95% confidence intervals; asterisks indicate Holm-adjusted $p < 0.05$ across all penalty-by-size comparisons. Positive differences favour the penalty. Zero-width intervals arise when all paired results coincide. This diagnostic is exploratory and does not replace the original H3 test.

Supplementary tables

Table S1. Full-array (16-electrode) leave-one-subject-out performance of three classifiers on GRABMyo (43 participants, 16 active gestures; chance Macro-F1 ≈ 0.063). CI: 95% participant cluster-bootstrap confidence interval. RBF-SVM: support vector machine with a radial basis function kernel. Training windows: windows used to fit each fold's model.

Model	Training windows	Macro-F1	95% CI	SD	Range
XGBoost	all (550,368 per fold)	0.542	0.517–0.567	0.085	0.385–0.702
RBF-SVM	40,000	0.538	0.513–0.562	0.085	0.383–0.718
Random forest	200,000	0.502	0.475–0.528	0.090	0.346–0.670

Paired comparisons (Wilcoxon signed-rank, Holm over 3): Random forest − RBF-SVM = −0.0362 (95% CI −0.0444 to −0.0284), $p_{\text{Holm}} < 0.001$, $r_{rb} = -0.97$; Random forest − XGBoost = −0.0408 (95% CI −0.0475 to −0.0340), $p_{\text{Holm}} < 0.001$, $r_{rb} = -0.98$; RBF-SVM − XGBoost = −0.0045 (95% CI −0.0107 to +0.0015), $p_{\text{Holm}} = 0.154$, $r_{rb} = -0.25$.

Table S2. Full-array XGBoost F1 score by active GRABMyo gesture (43 participants). Mean F1 across participants with a 95% percentile cluster-bootstrap confidence interval of the mean (10,000 resamples). Gesture names follow the order given in the dataset file MotionSequence.txt.

ID	Gesture	Mean F1	95% bootstrap CI
1	Lateral prehension	0.444	0.389 to 0.496
2	Thumb adduction	0.309	0.256 to 0.360
3	Thumb and little finger opposition	0.526	0.472 to 0.577
4	Thumb and index finger opposition	0.349	0.297 to 0.399
5	Thumb and index finger extension	0.486	0.424 to 0.547
6	Thumb and little finger extension	0.602	0.532 to 0.667
7	Index and middle finger extension	0.506	0.443 to 0.566
8	Little finger extension	0.499	0.441 to 0.553
9	Index finger extension	0.388	0.333 to 0.443
10	Thumb finger extension	0.538	0.470 to 0.603
11	Wrist extension	0.942	0.927 to 0.956
12	Wrist flexion	0.828	0.775 to 0.874
13	Forearm supination	0.814	0.770 to 0.852
14	Forearm pronation	0.491	0.431 to 0.550
15	Hand open	0.545	0.489 to 0.598
16	Hand close	0.410	0.368 to 0.449

Table S3. H1 decisions under multiplicity correction and sensitivity margins (GRABMyo, 43 participants, $\lambda = 0$). $\bar{\Delta}_k$ and lower 95% bound as in main table 1. p: one-sided cluster-bootstrap p for $H_0 : \bar{\Delta}_k \leq -\delta$ at the primary margin $\delta = 0.02$ (10,000 resamples); Holm over the 15 subset sizes at $\alpha = 0.025$. Fixed sequence (post-hoc sensitivity analysis): $k = 15, 14, \ldots$ tested in order, stopping at the first failure. The last two columns give the non-inferiority decision (unadjusted / Holm) at the prespecified sensitivity margins.

k	$\bar{\Delta}_k$	Lower bound	p	p_{Holm}	Non-inf.	Non-inf. (Holm)	Fixed sequence	$\delta = 0.01$	$\delta = 0.03$
15	−0.0036	−0.0056	<0.001	0.002	yes	yes	yes	yes / yes	yes / yes
14	−0.0060	−0.0081	<0.001	0.002	yes	yes	yes	yes / yes	yes / yes
13	−0.0109	−0.0133	<0.001	0.002	yes	yes	yes	no / no	yes / yes
12	−0.0133	−0.0160	<0.001	0.002	yes	yes	yes	no / no	yes / yes
11	−0.0204	−0.0246	0.579	1.000	no	no	no	no / no	yes / yes
10	−0.0259	−0.0311	0.984	1.000	no	no	no	no / no	no / no
9	−0.0318	−0.0387	0.999	1.000	no	no	no	no / no	no / no
8	−0.0366	−0.0440	1.000	1.000	no	no	no	no / no	no / no
7	−0.0413	−0.0489	1.000	1.000	no	no	no	no / no	no / no
6	−0.0610	−0.0713	1.000	1.000	no	no	no	no / no	no / no
5	−0.1133	−0.1277	1.000	1.000	no	no	no	no / no	no / no
4	−0.1444	−0.1590	1.000	1.000	no	no	no	no / no	no / no
3	−0.2015	−0.2212	1.000	1.000	no	no	no	no / no	no / no
2	−0.2420	−0.2603	1.000	1.000	no	no	no	no / no	no / no
1	−0.3603	−0.3832	1.000	1.000	no	no	no	no / no	no / no

k^* (unadjusted / Holm / fixed sequence) at $\lambda = 0$: $\delta = 0.01$: 14 / 14 / 14; $\delta = 0.02$: 12 / 12 / 12; $\delta = 0.03$: 11 / 11 / 11; identical for all four λ.

Table S4. Exploratory comparisons of SR-CFA ($\lambda = 0.5$) with other electrode-selection methods (GRABMyo, 43 participants; same comparison model as main table 2). Difference: SR-CFA minus comparator, mean paired difference with 95% participant cluster-bootstrap confidence interval (CI). p: Wilcoxon signed-rank, uncorrected; r_{rb}: matched-pairs rank-biserial correlation. These contrasts are exploratory and were not corrected for multiplicity.

k	Comparator	SR-CFA	Comparator	Difference	95% CI	p	r_{rb}
4	Permutation importance	0.383	0.385	−0.0018	−0.0151 to +0.0118	0.678	−0.08
	Geometric farthest-point	0.383	0.383	−0.0005	−0.0159 to +0.0143	0.986	+0.00
	Mutual information	0.383	0.280	+0.1031	+0.0880 to +0.1176	<0.001	+0.99
6	Permutation importance	0.453	0.440	+0.0126	+0.0047 to +0.0216	0.009	+0.52
	Geometric farthest-point	0.453	0.419	+0.0341	+0.0195 to +0.0476	<0.001	+0.68
	Mutual information	0.453	0.360	+0.0934	+0.0813 to +0.1050	<0.001	+1.00
8	Permutation importance	0.469	0.466	+0.0033	−0.0006 to +0.0071	0.069	+0.33
	Geometric farthest-point	0.469	0.465	+0.0047	−0.0044 to +0.0137	0.267	+0.20
	Mutual information	0.469	0.396	+0.0737	+0.0632 to +0.0841	<0.001	+1.00
12	Permutation importance	0.485	0.485	+0.0001	−0.0033 to +0.0033	0.995	−0.00
	Geometric farthest-point	0.485	0.478	+0.0070	+0.0000 to +0.0138	0.058	+0.33
	Mutual information	0.485	0.480	+0.0051	+0.0013 to +0.0089	0.007	+0.47

Table S5. H4: deletion with retraining (GRABMyo, 43 participants). Retained: electrodes kept after deletion. Remove-Top, Remove-Bottom and Remove-Random delete the highest-ranked, lowest-ranked or randomly chosen electrodes; Remove-Random uses the random subsets of the H2 comparison and exists only where these were sampled. All orders use the same comparison model and training sample. Difference: mean paired difference (first minus second order) with 95% participant cluster-bootstrap confidence interval (CI); negative values follow the expected ordering. p_{Holm}: Wilcoxon signed-rank, Holm over the 15 contrasts; r_{rb}: matched-pairs rank-biserial correlation.

Retained	Contrast	First	Second	Difference	95% CI	p_{Holm}	r_{rb}
2	Top − Bottom	0.263	0.292	−0.0297	−0.0419 to −0.0166	<0.001	−0.67
4	Top − Bottom	0.351	0.382	−0.0314	−0.0447 to −0.0177	<0.001	−0.64
	Top − Random	0.351	0.366	−0.0151	−0.0239 to −0.0061	0.010	−0.51
	Random − Bottom	0.366	0.382	−0.0163	−0.0265 to −0.0061	0.013	−0.47
6	Top − Bottom	0.376	0.454	−0.0775	−0.0921 to −0.0627	<0.001	−0.96
	Top − Random	0.376	0.416	−0.0398	−0.0487 to −0.0307	<0.001	−0.91
	Random − Bottom	0.416	0.454	−0.0377	−0.0459 to −0.0293	<0.001	−0.93
8	Top − Bottom	0.408	0.469	−0.0618	−0.0772 to −0.0461	<0.001	−0.89
	Top − Random	0.408	0.445	−0.0377	−0.0476 to −0.0273	<0.001	−0.86
	Random − Bottom	0.445	0.469	−0.0242	−0.0309 to −0.0172	<0.001	−0.86
10	Top − Bottom	0.445	0.475	−0.0297	−0.0399 to −0.0188	<0.001	−0.73
12	Top − Bottom	0.455	0.485	−0.0302	−0.0385 to −0.0218	<0.001	−0.84
	Top − Random	0.455	0.478	−0.0228	−0.0296 to −0.0158	<0.001	−0.81
	Random − Bottom	0.478	0.485	−0.0074	−0.0101 to −0.0046	<0.001	−0.74
14	Top − Bottom	0.487	0.491	−0.0038	−0.0084 to +0.0009	0.150	−0.25

Table S6. H3: stability-regularised ranking ($\lambda > 0$) minus magnitude-only ranking ($\lambda = 0$) on the electrode-count performance curve (GRABMyo, 43 participants). Mean paired difference with 95% participant cluster-bootstrap confidence interval (CI). Tied: participants for whom both rankings select the identical subset. p_{Holm}: Wilcoxon signed-rank, Holm over 12 tests. r_{rb} is computed from untied participants only and is undefined (–) when all are tied; with most participants tied it is not informative about magnitude.

λ	k	Mean difference	95% CI	Tied	p_{Holm}	r_{rb}
0.25	4	−0.0002	−0.0005 to +0.0000	41/43	1.000	−1.00
0.25	6	+0.0000	+0.0000 to +0.0000	43/43	1.000	–
0.25	8	+0.0001	+0.0000 to +0.0002	42/43	1.000	+1.00
0.25	12	+0.0000	+0.0000 to +0.0000	43/43	1.000	–
0.5	4	+0.0004	−0.0005 to +0.0018	40/43	1.000	+0.00
0.5	6	+0.0000	+0.0000 to +0.0000	43/43	1.000	–
0.5	8	+0.0008	+0.0000 to +0.0019	40/43	1.000	+1.00
0.5	12	+0.0000	+0.0000 to +0.0000	43/43	1.000	–
1	4	+0.0010	−0.0003 to +0.0027	38/43	1.000	+0.47
1	6	+0.0000	+0.0000 to +0.0000	43/43	1.000	–
1	8	+0.0015	+0.0004 to +0.0028	36/43	0.216	+1.00
1	12	−0.0005	−0.0014 to +0.0000	42/43	1.000	−1.00

Table S7. Participant-level attribution extension (exploratory; GRABMyo). Non-inferiority uses the same-budget full-array reference and the original engineering margin. The paired difference compares participant versus fold aggregation at the same penalty and electrode count.

λ	k^*	k^* (Holm)	Δ at $k = 12$ (95% CI)	p_{Holm}
0	12	12	+0.0000 (+0.0000, +0.0000)	1.000
0.25	12	12	+0.0000 (+0.0000, +0.0000)	1.000
0.5	12	12	+0.0000 (+0.0000, +0.0000)	1.000
1	12	12	+0.0005 (+0.0000, +0.0014)	1.000

Table S8. NinaPro DB5 channel-count performance curve (exploratory replication; cross-fitted ranking, $\lambda = 0$; 10 subjects; reference $k = 16$, mean Macro-F1 0.281; margin $\delta = 0.02$). The lower 95% bound is shown to six decimals because the $k = 12$ decision lies at the margin. Non-inferior: lower bound above $-\delta$, unadjusted and after Holm correction over 15 subset sizes. Bootstrap reruns: number of 200 reruns of the bootstrap with different random seeds in which the unadjusted decision was non-inferior. This is a Monte Carlo decision-stability diagnostic, not a probability that the hypothesis is true.

k	Macro-F1	Δ_k	Lower bound	Non-inf.	Non-inf. (Holm)	Non-inf. in bootstrap reruns
15	0.270	−0.0108	−0.023773	no	no	0/200 (0.0%)
14	0.269	−0.0119	−0.027438	no	no	0/200 (0.0%)
13	0.266	−0.0143	−0.031186	no	no	0/200 (0.0%)
12	0.268	−0.0126	−0.019994	yes	no	191/200 (95.5%)
11	0.260	−0.0211	−0.034121	no	no	0/200 (0.0%)
10	0.265	−0.0160	−0.031368	no	no	0/200 (0.0%)
9	0.260	−0.0202	−0.036943	no	no	0/200 (0.0%)
8	0.252	−0.0288	−0.055369	no	no	0/200 (0.0%)
7	0.232	−0.0486	−0.075346	no	no	0/200 (0.0%)
6	0.231	−0.0498	−0.081816	no	no	0/200 (0.0%)
5	0.200	−0.0803	−0.118422	no	no	0/200 (0.0%)
4	0.180	−0.1006	−0.150266	no	no	0/200 (0.0%)
3	0.162	−0.1185	−0.174145	no	no	0/200 (0.0%)
2	0.138	−0.1430	−0.195307	no	no	0/200 (0.0%)
1	0.105	−0.1754	−0.229548	no	no	0/200 (0.0%)

Table S9. Cross-session evaluation on GRABMyo (exploratory; full array). All conditions are also subject-independent: the test participant is never in training.

Design	Train sessions	Test session	Train windows	Macro-F1
CS1	1	1	183,456	0.492 ± 0.101
CS1	1	2	183,456	0.528 ± 0.088
CS1	1	3	183,456	0.510 ± 0.090
CS2	1+2	3	366,912	0.530 ± 0.095

Paired contrasts (exploratory, uncorrected): CS1: test S2 − test S1: +0.0355 (95% CI +0.0091 to +0.0621), p = 0.024; CS1: test S3 − test S1: +0.0172 (95% CI −0.0038 to +0.0391), $p = 0.272$; S3 test: CS2 (train S1+S2) − CS1 (train S1): +0.0210 (95% CI +0.0146 to +0.0274), $p < 0.001$.

Table S10. GRABMyo derived bipolar pairs $F_j - F_{j+8}$ (exploratory; 43 participants; $\lambda = 0$; count-feature threshold 4×10^{-3} mV; margin $\delta = 0.02$). Reference: the $k = 8$ model of the same curve, trained on the curve's 200,000-window sample (mean Macro-F1 0.480). The full-array bipolar value reported in the main text (0.478) comes from a separate model trained on all 550,368 training windows of each fold, as for the monopolar analysis. Δ_k: mean paired difference with 95% participant cluster-bootstrap confidence interval (CI). Bootstrap reruns: as in supplementary table S8.

k	Macro-F1	Δ_k	95% CI	Non-inf.	Non-inf. in bootstrap reruns
7	0.464	−0.0158	−0.0203 to −0.0113	no	0/200 (0.0%)
6	0.442	−0.0375	−0.0451 to −0.0303	no	0/200 (0.0%)
5	0.405	−0.0742	−0.0845 to −0.0646	no	0/200 (0.0%)
4	0.356	−0.1240	−0.1392 to −0.1096	no	0/200 (0.0%)
3	0.310	−0.1697	−0.1852 to −0.1546	no	0/200 (0.0%)
2	0.240	−0.2399	−0.2615 to −0.2177	no	0/200 (0.0%)
1	0.174	−0.3052	−0.3298 to −0.2790	no	0/200 (0.0%)